

A Study on Scalable Parallelism in Spider-Web Networks

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Abstract

Regular degree-three Spider-Web networks, $SW(m,n)$, are optimally, scalability prototyped along transportation paths as dual-surveillance – preventing information occlusion, or reliable identity based telecommunication networks. To promote network information transmission such as coping with radio interference and multipath effects, offering dynamic authentication / authorization the performance of mutually independent Hamiltonian paths or parallelism is considered. Specifically, for any pair of bipartite nodes, such networks can offer at least two parallel computing paths efficiently, extensively and functionally pass through all other nodes. A proof methodology, with ideas like incorporated alternating embeddings and morphological extensions, is generated to present the parallelism inherited in this kind of network.

Keywords: mutually-independent Hamiltonian paths (MIHP); parallel computing; transportation

1 Introduction

Offering real-time location-based information wirelessly is essential for business development or human welfare in the contemporary information era. Safety or security is the highest or nonnegotiable welfare generally. In critical locations, where the line-of-sight of tracked objects, vehicles, or people being occluded in random conditions such as traffic congestions can be a concern [Hsu 2008]. Similar to the impact of the line-of-sight interference, the non-line-of-sight wireless communication can also easily be interfered (e.g., due to vehicles' metal material in transportation environment) or be attacked to affect privacy, safety, or other qualities [Chen et. al. 2009; Hsu 2009a].

Sensors being incorporated can provide more functionality by differential sensing, comparison, and adaptation (e.g., incorporated cameras adaptively applied to accommodate varying light environments). The cost of sensor installment for critical locations is becoming less essential, especially compared to the price of casualty

impacts - e.g., that occurred at a dominant avenue of the capital, even its probability may be considered small by most civilians. Basically, such costs of sensor installment can become cheaper with technological advancement, in other words, it can be more affordable yet with more functionality.

The supervisory control and data acquisition (SCADA) network with parallel surveillance or adaptable detection availability is considered along critical passages. The spider-web network, $SW(m,n)$, is hereby prototyped. It is scalable -conforming to both the width and the length of the path, or conforming to the level of service. The degree (i.e., links to each node) of nodes of $SW(m,n)$ is regularly optimized three, in terms of considering basic parallelism (dual-surveillance) and practical working order for maintenance operations in only one lane of the path. Such parallelism can allow for more security features than the traditional ring network can.

Visibly, just as drivers' necessary concerning both-sides' views along moving activities with their two eyes, it is worthwhile to help communicate with the human or the vehicle on critical passages through well receiving information related with three dimensional (or both-sides') real-time factors. Invisibly, wireless communication data, interfered or attacked, can be better identified by comparing the difference among signals acquired through parallel sequences of location defined sensors [i.e., the performance of mutually independent Hamiltonian paths (MIHP)]. In section 3 after mathematical reviews in section 2, advanced morphing studies on MIHP will be developed, to be shown as the proof examples in section 4. Finally, we have some conclusions.

2 Mathematical Preliminaries

Communication networks are usually illustrated by graphs in which nodes represent processors and edges represent links between processors. Let $G=(V,E)$ be a graph if V is a finite set and E is a subset of $\{(a,b) | (a,b) \text{ is an unordered pair of } V\}$. A path is delimited by $(x_0, x_1, x_2, \dots, x_{n-1})$. A path is called a Hamiltonian path if its nodes are distinct

and span V . A cycle is a path of at least three nodes such that the first node is the same as the last node. A cycle is called a Hamiltonian cycle or Hamiltonian if its nodes are distinct except for the first node and the last node, and if they span V .

A bipartite graph $G = (V, E)$ is a graph such that $V = A \cup B$ and E is a subset of $\{(a, b) \mid a \in A \text{ and } b \in B\}$; if $G - F$ remains Hamiltonian for any $F = \{a, b\}$ with $a \in A$ and $b \in B$, then G is I_p -Hamiltonian. A graph G is I -edge Hamiltonian if $G - e$ is Hamiltonian for any $e \in E$; moreover, if there is a Hamiltonian path between any pair of nodes $\{c, d\}$ with $c \in A$ and $d \in B$, then the bipartite graph G is Hamiltonian laceable.

The bipartite Spider-Web network, $SW(m, n)$, is a graph with the node set $\{(i, j) \mid 0 \leq i < m, 0 \leq j < n\}$, where m and n are ≥ 4 , even integers such that (i, j) and (k, l) are adjacent if they satisfy one of the following conditions: (1) $i = k$ and $j = l \pm 1$; (2) $j = l$ and $k = i + 1 \pmod{m}$ (also presented as $k = [i + 1]_m$) if $i + j$ is odd or $j = n - 1$; (3) $j = l$, $k = i - 1 \pmod{m}$ if $i + j$ is even or $j = 0$. $SW(m, n)$ is proved to be I -edge Hamiltonian and I_p -Hamiltonian [Kao and Hsu 2005a]. Thus, the fault-tolerance engaged in is systematically based. Moreover, $SW(m, n)$ are Hamiltonian laceable [Kao and Hsu 2005b].

The number of links connecting a node is called the degree; networks regularly have fewer degrees and are generally economic [Stojmenovic 1997]. Two Hamiltonian paths $P_1 = (u_1, u_2, \dots, u_{n(G)})$ and $P_2 = (v_1, v_2, \dots, v_{n(G)})$ of G from u to v are independent if $u = u_1 = v_1$, $v = u_{n(G)} = v_{n(G)}$, and $u_i \neq v_i$ for every $1 < i < n(G)$. A set of Hamiltonian paths, $\{P_1, P_2, \dots, P_k\}$, of G from u to v are mutually independent if any two distinct paths in the set are independent from u to v . It was found that $SW(m, n)$ has the performance of at least two mutually independent Hamiltonian paths between any pair of bipartite nodes.

3 Morphing MIHP on Spider-Web Networks

Based on the definition of MIHP, we can assume that u_i, v_i are the sequential order numbers for a node of a pair of Hamiltonian-path patterns, P_1 and P_2 respectively. The patterned $SW(m, n)$ can be extended by two basic means: linear extension and helix extension. The Hamiltonian path has two end-nodes (x_1, y_1) , (x_2, y_2) , $y_1 \geq y_2$. The linear extension only overlaps at the part above y_1 and below y_2 ; the helix extension is only approached in the part between y_1 and y_2 .

Let a and b be the linear extension of above y_1 or below y_2 respectively, where a or b can be any even non-negative integer. The upper extension, a , is from w to x . The lower extension, b , is from y to

z . $SW(4, n)$ has four columns; these column indices can be represented by A, B, C, D respectively. For example, in column D , the upper linear extension can be presented as the extensive block $Dw-Dx$ or $Dx-Dw$ [Hsu 2009b], depending on the sequence direction. Let Dw, Dx or similar codes, which have half number of nodes in a block, be the extensive cluster. An extensive cluster can generally be defined as a special node of various numbers of sub-nodes. On illustrations, such a pair of special nodes may be disconnected, where one may contain nodes of two clusters (of a block) presented as DW or DX (named as l -cluster or s -block), and the other one contains only one node of the named cluster but still presented as Dx or Dw .

The helix extension path passes through the two nodes of every column in a patterned cyclic path which may be presented as, $D-v-D$, where $-v-$ represents a helix extension and D is the index of the start/end column. In a helix extension, each of the two nodes of paired columns may be adaptively moved to different elevations, or more nodes may be added in its first or last extension. It is essential that nodes of $SW(m, n)$ have only one link extending horizontally so the sequences collision can easily be perceived.

In $SW(4, n)$, $0 \leq \alpha_{i1}, \alpha_{i2}, \beta_{i1}, \beta_{i2} \leq 4$: α_{i1} (α_{i2}) is the number of upper-extensive blocks before node i in P_{1e} (P_{2e}), or extension patterns of P_1 (P_2), while β_{i1} or β_{i2} is coded for the lower extensive blocks.

Let H_r or H_l be the number of horizontal extensions in the location-right or location-left part to keep the original lead/lag relationship. For example, if $u_i < v_i$, then it is hoped that $(u_i + \alpha_{i1} * a + \beta_{i1} * b + \sum_{s=1}^{s=s1-1} \gamma_{i1s} * H_r + \sum_{t=1}^{t=t1-1} \lambda_{i1t} * H_l) < (v_i + \alpha_{i2} * a + \beta_{i2} * b + \sum_{s=1}^{s=s2-1} \gamma_{i2s} * H_r + \sum_{t=1}^{t=t2-1} \lambda_{i2t} * H_l)$ [Hsu 2009b].

$\sum_{s=1}^{s=s1-1} \gamma_{i1s} (\sum_{t=1}^{t=t2-1} \gamma_{i2t})$ represents the total number of nodes after the location-right insertion on the edges which affects the sequence of node i in path P_1 (P_2); the code γ represents the embedding with right insertion. Similarly, $\sum_{t=1}^{t=t1-1} \lambda_{i1t} (\sum_{s=1}^{s=s2-1} \lambda_{i2s})$ represents the total number of nodes after the location-left insertion on the edges which affects the sequence of node i in path P_1 (P_2); the code λ represents the embedding with left insertion. The possibility of linear extension in horizontal embeddings needs to be considered.

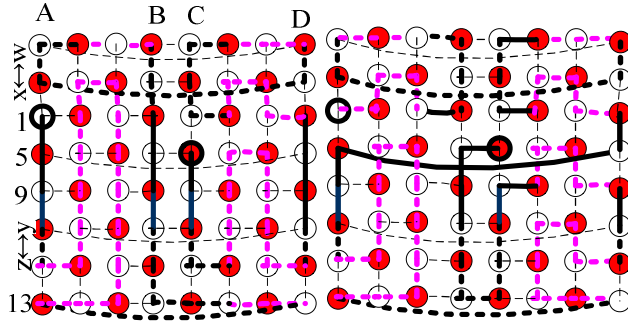
We can also check whether the separation of the sequence is a number which cannot be equaled by a constantly increasing means (i.e., if the separation is six and the constantly increasing means is the multiple of four, six cannot be equal to four or its multiples).

MIHP properties are studied in the even-/odd-column- separation and at-the-same-column classifications. The first classification is

introduced in this paper, but the rest classifications are only illustrated in the appendix. It was found that such performance can be simply established through vertical extensions, horizontal extensions, incorporated alternating embeddings and morphology configurations.

4 Proof Examples ($x_2 - x_1 \neq \text{even}$)

A. $y_1 - y_2 = 1$; ($y_1 \leq n-1, y_2 \geq 2$) or ($y_1 \leq n-3, y_2 \geq 0$)



The below two basic Lt. and Rt. Hamiltonian path patterns in a SW(4,4) network apparently have no collision except for end nodes.

01,05,09,13,16,15,14,10,06,02,03,04,08,12,11,07 Lt.
01,02,03,04,08,05,09,13,16,12,11,15,14,10,06,07 Rt.

(1). Basic patterns can be definitely extended downward, i.e., $b(y \leftrightarrow z) \neq 0$ (Figure 1).

Figure 1. Two SW(m,n) MIHP fm. (x_1, y_1) to (x_2, y_2); $x_1 - x_2$ even, $y_1 - y_2 = 1$; $y_1 \leq n-1, y_2 \geq 2$

Nodes 05, 09, 13, 14, 10, 06 and clusters Ay, Az, By, Bz of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 02, 03, 04, 08, 16, 12, 11, 15 and clusters Ax, Bx, Cx, Dx, Bw, Aw, Dw, Cw of Rt. pattern keep lag with definitely not-more adding nodes. Embedded nodes (including cluster nodes) within 01, 02 and Ay, 13 in Rt. pattern keep same node sequence and definitely node separations with Lt. pattern in sections.

Compared with Rt. pattern's 16, Dz, in Lt. pattern, its embedding clusters which are within Dz, Cz keep lead with potentially more nodes in

front. Similarly, in Lt. pattern, the embedding nodes within 16, 15 keep lag with those within Rt. pattern's 16, Dz. Then in Rt. pattern's 12, 11, embedding nodes do not collide due to having consistent order with those within Dz, Cz in Lt. pattern and with same total embedding nodes (including clusters) after the essentially first embedding. Rt. pattern's 16, Dz embedding needs $b \neq 0$.

(2). Basic patterns can be only extended upward, i.e., $b(y \leftrightarrow z) = 0$ (Figure 2).

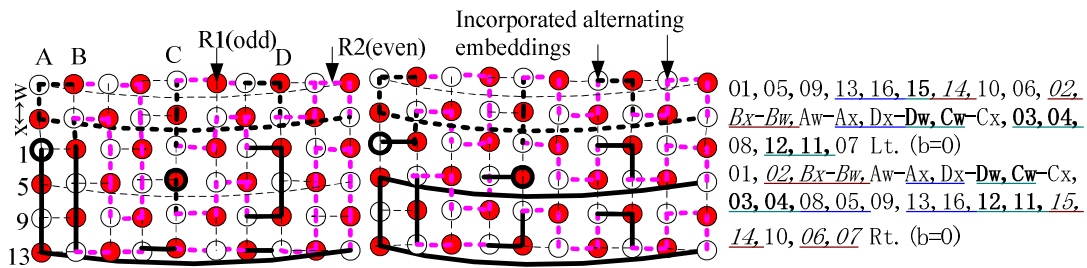


Figure 2. Two SW(m,n) MIHP fm. (x_1, y_1) to (x_2, y_2); $x_1 - x_2$ even, $y_1 - y_2 = 1$; ($y_1 \leq n-1, y_2 = 2$) or ($y_1 = n-3, y_2 \geq 0$)

Nodes 05, 09, 13, 16, 15, 14, 10, 06 of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 02, 03, 04, 08, 12, 11 and clusters Aw, Ax, Dx, Bx, Bw, Dw, Cw, Cx of Rt. pattern keep lag with definitely not-more adding nodes. 08, 05 of Rt. pattern has no collision with 13, 16 of

Lt. pattern by 03, 04 alternating incorporation. Nodes within 13, 16 of Rt. pattern keep lead of the same nodes of Lt. pattern by definitely not-less adding nodes.

Embedded nodes within Dw, Cw, 03, 04, 12, 11, in Rt. pattern are not wholly same as those in Lt.

pattern; however, by incorporated embeddings, they will still keep lag with corresponding yet crossing sections in Lt. pattern, including 16,15, respectively.

Embedding nodes within Bx-Bw in Rt. pattern keep lag by less adding nodes with those corresponding nodes within Bx-Bw in Lt. pattern. If a $(w \leftrightarrow x)=0$, in both patterns, 02,03 will replace Bx-Bw, which combines a set of pulse embeddings (imaginary Bx-Cx) and a set or row embeddings (imaginary Cw-Bw). Embedding nodes within 06,07, in Rt. pattern naturally keep lag by less adding nodes with those corresponding nodes within Bx-Bw in Lt. pattern by more adding nodes after the essential check of the first embedding.

B. $y_1-y_2=1, y_1 \leq n-2, y_2 \geq 1$ (Figure 3)

The below two basic Lt. and Rt. Hamiltonian path patterns in a SW(4,4) network apparently have no collision except for end nodes.
05,01,02,06,10,09,13,14,15,16,12,08,04,03,07,11 Lt.

05,08,04,01,02,03,07,06,10,09,13,14,15,16,12,11 Rt.

Network configurations, $y_1 \leq n-2, y_2 \geq 1, y_1-y_2=1$, can keep same if we turn them upside down and change column codes from A,B,C,D to A,D,C,B respectively. Then we can compare the original paired pattern yet codes A,B,C,D changed to C,D,A,B respectively, and can find that assuming $R-emb \geq L-emb$ is reasonable.

Nodes 01, 02, 06, 10, 09, 13, 14, 15, 16, 12 and clusters Ax, Aw, Bw, Bx, Ay, Az, Bz, By,Cy, Cz, Dz, Dy of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 08, 04, 03, 07 and clusters Cw, Cx, Dw, Dx of Rt. pattern keep lag with definitely not-more adding nodes. Nodes within 02,03 and 07,06 of Rt. pattern keep lead with definitely more adding nodes. Embedding nodes, 6 nodes and 2 lower clusters per embedding, within 05,08 of Rt. pattern keep lag of the same nodes of Lt. pattern's Dw,04 (with 8 nodes, 2 lower clusters and 2 upper clusters), by definitely less adding nodes. Similarly, embedding nodes within 04,01 of Rt. pattern keep lag of the corresponding nodes of Dw,04 in Lt. pattern.

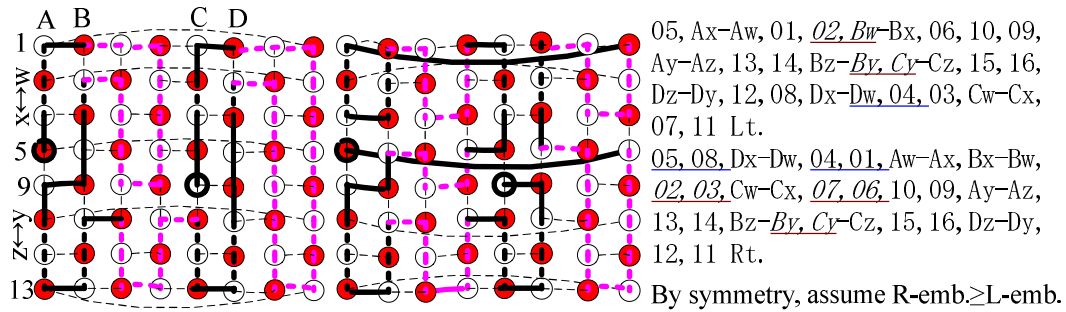


Figure 3. Two SW(m,n) MIHP fm. (x_1, y_1) to (x_2, y_2) ; x_1-x_2 even, $y_1-y_2=1, y_1 \leq n-2, y_2 \geq 1$

C. $y_1 \leq n-2, y_2 \geq 1, y_1-y_2 \geq 3$ (Figure 4)

The below two basic Lt. and Rt. Hamiltonian path patterns in a SW(4,6) network apparently have no collision except for end nodes.
05,01,04,08,12,11,07,03,02,06,10,09,13,16,20,24,23,22,21,17,18,14,15,19 Lt.
05,09,10,06,07,03,02,01,04,08,12,11,15,14,18,22,21,17,13,16,20,24,23,19 Rt.

Both Lt. and Rt. patterns can be separated by the helix extension, which starts from different column. Both patterns have same number of nodes in the upper and the lower part. The network degree is regular three, only one horizontal link links a node, and their helix extensions are essentially same. Therefore, these two helix extensions of both patterns do not collide; their sequentially ordered nodes are naturally different. Similar to case: $y_1 \leq n-2, y_2 \geq 1, y_1-y_2=1$; we

assume $R-emb \geq L-emb$.

On the upper part, nodes 01,04, 08, 12, 11 and clusters Ax-Aw, Dw-Dx of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 09, 10, 06, 07, 03, 02 and clusters Cx-Cw, Bw-Bx of Rt. pattern keep lag with definitely not-more adding nodes. Embedded nodes (including cluster nodes) within 06,07 and 03,02 in Rt. pattern can keep lag with those within 03,02 in Lt. pattern due to $R-emb \geq L-emb$.

On the lower part, nodes 13,16, 20, 24, 23 and clusters Dy, Dz, Cz, Cy of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 15, 14, 18, 22, 21, 17 and clusters By, Bz, Az, Ay of Rt. pattern keep lag with definitely not-more adding nodes. Nodes within 15,14, (including embedded cluster nodes) in Rt. pattern can keep

lag with those in Cy,By and 14,15 in Lt. pattern

due to $R\text{-emb.} \geq L\text{-emb.}$

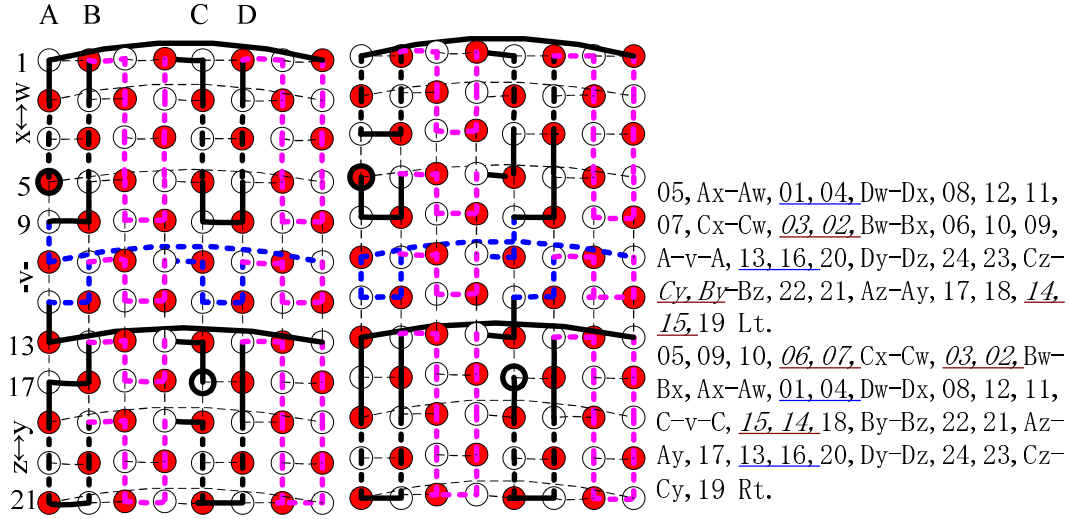


Figure 4. Two SW(m,n) MIHP fm. (x1,y1) to (x2,y2); x_1-x_2 even, $y_1 \leq n-2$, $y_2 \geq 1$, $y_1-y_2 \geq 3$

D. $y_1=n-1$, $y_2=0$ (Figure 5)

The below two basic Lt. and Rt. Hamiltonian path patterns in a SW(4,4) network apparently have no collision except for end nodes.
 01,04,03,02,06,07,11,12,08,05,09,10,14,13,16,15 Lt.
 01,05,08,04,03,02,06,07,11,12,16,13,09,10,14,15 Rt.

Similar to the case: $y_1 \leq n-2$, $y_2 \geq 1$, $y_1-y_2=1$; we assume $L\text{-emb.} \geq R\text{-emb.}$ In Rt. pattern, nodes 05, 08, 16, 13 keep lag with definitely not-more adding nodes; nodes 04, 11, 12, 09, 10, 14, 03,02, and nodes of 06,07 when the helix extension exists, can keep lead with definitely not-less adding nodes. By helix extensions which start from

different columns - column C and D as the figure, both Lt. and Rt. patterns can be separated into the upper and the lower parts having equal number of nodes. The network's degree is regular three, so only one horizontal link links a node. Besides, helix extensions basically have the alternating "vertical", "horizontal" order. Therefore, this mechanism can help that the LT-Rt. paired helix extensions do not collide with each other.

$L\text{-emb.} \geq R\text{-emb.}$ can help nodes' lead/lag relationship kept - including nodes within 14,15 of Rt. Pattern when the helix extension exists, and within 06,07 of Lt. pattern when no helix extension exists.

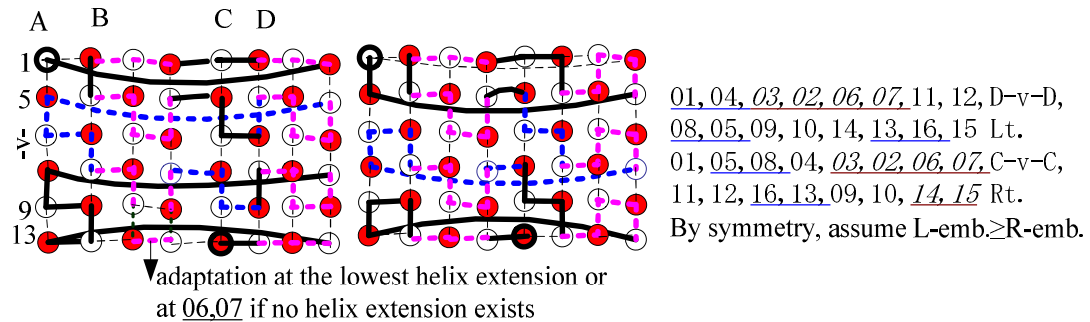


Figure 5. Two SW(m,n) MIHP fm. (x1,y1) to (x2,y2); x_1-x_2 even, $y_1=n-1$, $y_2=0$

E. $y_1-y_2 \geq 3$; ($y_1 \leq n-3$, $y_2 \geq 0$) or ($y_1 \leq n-1$, $y_2 \geq 2$)

The below two basic Lt. and Rt. Hamiltonian path patterns in a SW(4,6) network apparently

have no collision except for end nodes.

09,13,16,12,11,07,03,04,08,05,01,02,06,10,14,15, 19,20,24,21,17,18,22,23 Lt.
 09,05,01,04,08,12,11,07,03,02,06,10,14,15,19,20,

16,13,17,18,22,21,24,23 Rt.

Both Lt. and Rt. patterns can be separated by the helix extension, which starts from same column yet under lead/lag order, and these two helix extensions of both patterns do not collide because their sequentially ordered nodes are naturally different. Rt. pattern has same number of nodes in the upper and the lower part; Lt. pattern

has four more number of nodes in the upper part.

(1). Basic patterns can be definitely extended downward, i.e., $b(y \leftrightarrow z) \neq 0$ (Figure 6).

Similar to case- $(y_1 \leq n-2, y_2 \geq 1, y_1 - y_2 \geq 3)$, we can assume: $R\text{-emb} \geq L\text{-emb}$.

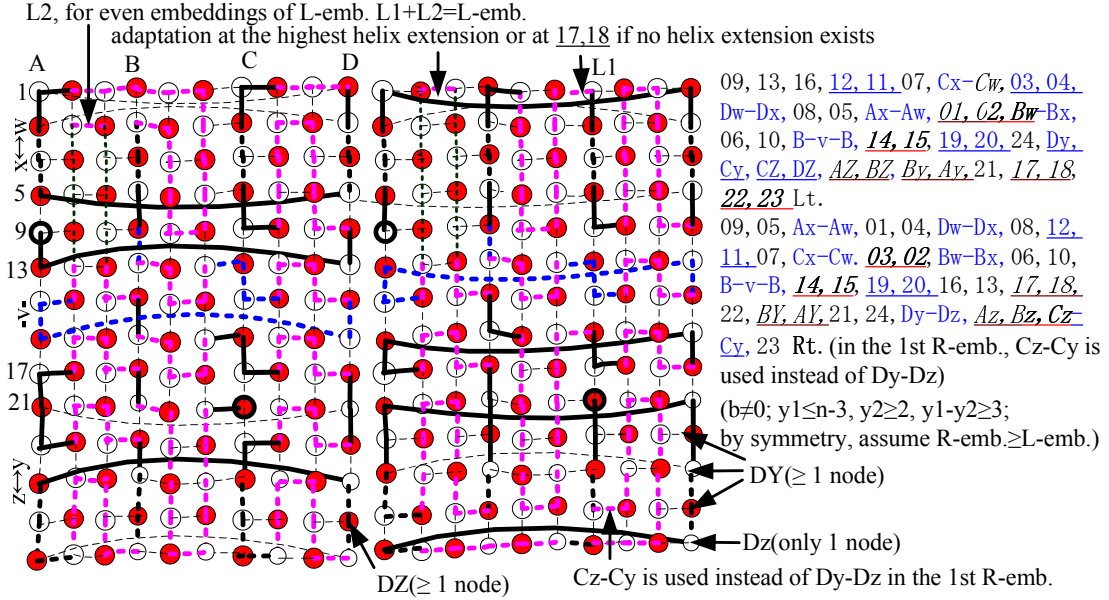


Figure 6. Two SW(m,n) MIHP fm. (x_1, y_1) to (x_2, y_2) ; $x_1 - x_2$ even, $y_1 - y_2 \geq 3$; $y_1 \leq n-3, y_2 \geq 2$

On the upper part, nodes 13, 16, 12, 11, 07, and clusters Cx, Cw of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 03, 04, 08, 05, 01, 02, 06, 10 and clusters Dw, Dx, Ax, Aw, Bw, Bx of Lt. pattern keep lead with definitely not-less adding nodes. Nodes (including node clusters) within 02, Bw of Lt. pattern are combined by nodes (including node clusters) within (imaginary) row embeddings- 02, 03 and within (imaginary) purse embeddings- Cw, Bw, which have consistent node sequential order as purse embeddings- 03, 02 of Rt. pattern. Hence, once the first embedding being checked collision free then such performance will be kept; similarly, the adaptation part of the helix extension (total adding nodes of both patterns are same) can be checked collision free.

On the lower part, nodes 14, 21, 17, 18, 22 of Rt. pattern keep lag with definitely not-more adding nodes; node 24, node 15, node Dy (of Dy, Dz), and 19, 20 after the first L-emb., and Az, Bz of Rt. pattern, keep lead with definitely not-less adding nodes. However nodes within Bz, Cz in Rt. pattern keep lag with compared nodes within 22, 23 of Lt. pattern due to Cz-Cy in Rt. pattern being inserted in the first R-emb.

L-emb. can include incorporated alternating

embeddings L1 (for odd embeddings of L-emb.) and L2 (for even embeddings of L-emb.). Before the numbered R-embedding equaling L-emb., incorporated alternating embeddings can help that DZ (contains some Dy) of Lt. pattern cannot touch Dy of Rt. pattern. Specifically, in the Rt. pattern, that L1's more inserting nodes within 14, 15 (2b or 2 less nodes than nodes within Dy, Cy and Cz, Dz) plus L2's inserting nodes within 17, 18, By, Ay (2 more nodes than nodes within Dy, Cy and Cz, Dz of Lt. pattern) can equalize two R-emb's inserting nodes within Dy, Cy and Cz, Dz. More specifically, (BY, AY, AZ, or BZ contains b-1 nodes) such incorporated alternating embeddings help AZ of Lt. pattern keep lag with AY of Rt. pattern. We can assign Dy-Dz of Rt. pattern being Dy and Dz, which can be considered as a special node or a mass similar to DZ of Lt. pattern but upside-down. For concerning the collision between Bz, Cz of Rt. pattern and 22, 23 of Lt. pattern, Cz-Cy of Rt. pattern is the first R-emb. Depending on L-emb. being even or odd, the net separation between both DZ is 3 or 5 nodes. However, the per-embedding nodes within Dy, Cy and Cz, Dz are $2b+2 \geq 6$ nodes. Hence, both DZ will not collide. Naturally, inserting nodes within Cz-Cy cannot be collided because Dy-Dz has $2b+2$ nodes itself being as a

buffer to protect $Cz-Cy$.

(2). Basic patterns cannot be extended downward, i.e., $b(y \leftrightarrow z)=0$ (Figure 7).

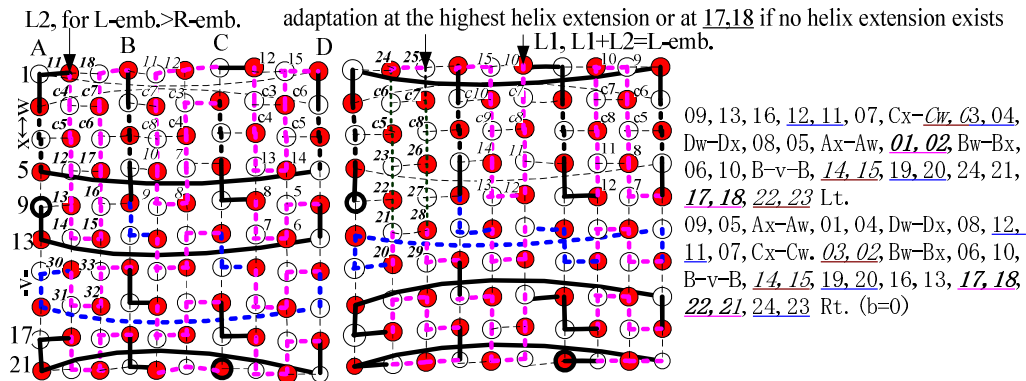


Figure 7. Two SW(m,n) MIHP fm. (x1,y1) to (x2,y2); x1-x2 even, y1-y2 ≥ 3; (y1 ≤ n-3, y2=0) or (y1=n-1, y2 ≥ 2)

On the upper part, nodes 13, 16, 12, 11, 07, and clusters Cx, Cw of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 01, 02, 03, 04, 08, 05, 06, 10 and clusters Dw, Dx, Ax, Aw, Bw, Bx of Lt. pattern keep lead with definitely not-less adding nodes. Nodes (including node clusters) within Cw_{03} (or 07_{03}) of Lt. pattern are combined by nodes (including node clusters) within (imaginary) row embeddings- 02_{03} and within (imaginary) purse embeddings- Cw_{Bw} (or 07_{06}), which have consistent node sequential order as purse embeddings- 03_{02} of Rt. pattern. By basic lead/lag relationships, within-nodes per embedding and total nodes per embedding, once the first embedding being checked collision-free at the worst condition then such performance will be kept.

On the lower part, the existence of nodes within 17_{18} or 22_{21} need $L1=R-emb.$; hence nodes 14, 15, 19, 20, 17, 18, 22 of Rt. pattern keep lag with definitely not-more adding nodes, and node 21, 24 keep lead with definitely not-less adding nodes. The only different quantity situation of nodes-within mark in both patterns is 14_{15} , which has already been discussed.

5 Conclusion

Mathematical spider-web networks, SW(m,n) are prototyped as identity-oriented SCADA networks along critical paths. Especially concerning on that wireless communication data, interfered or attacked, can be better identified by comparing the difference among signals acquired through parallel sequences of location defined sensors, so that the performance of mutually independent Hamiltonian paths (MIHP) is studied. It is found such performance can be established with simplicity by vertical extensions, horizontal

extensions, and morphology configurations.

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Appendix

A. MIHP properties of the odd-column-separation classification



Figure 8. MIHP Properties in the Odd Column Separation for the Spider-Web Networks.

Proof on subcase (c): $y_1 - y_2 \geq 2$, $x_1 = 0$, x_2 odd, $(y_1 = n-1, y_2 = 1)$ or $(y_1 = n-2, y_2 = 0)$

(1). Consider $[x_2 > 1; y_1 - y_2 \geq 4]$ and $[y_1 - y_2 \geq 2; (x_2 = 1, y_1 = n-1, y_2 = 1)$ or $(m - x_2 = 1, y_1 = n-2, y_2 = 0)]$.

The below two basic Lt. and Rt. Hamiltonian path patterns in a SW(4,4) network apparently have no collision except for end nodes.

01,04,03,02,06,07,11,12,08,05,09,13,16,15,14,10 Lt.

01,05,08,04,03,02,06,07,11,12,16,15,14,13,09,10 Rt.-(L-emb=0)

The third basic pattern as below is used for L-emb.>0 (as the italic position) and needs $y_1 - y_2 \geq 4$:

01,05,08,04,03,02,06,07,12,16,13,09,11,15,14,10 Rt.-(L-emb.>0)

If L-emb.=0 and $y_1 - y_2 \geq 2$, the helix extensions of both patterns start at the same location and have same vertical-horizontal sequence, but at different sequential order, hence they will not collide. On the upper part, the Lt. pattern has 6 nodes, and the Rt. pattern has 8 nodes. Nodes 04, 03, 02, 06, 07, 11, 12, 09, 13 of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 08, 05, 16, 15, 14 of Lt. pattern keep lead with definitely not-less adding nodes.

If L-emb. $\neq 0$, we consider $y_1 - y_2 \geq 4$. Lt. pattern needs a required helix extension; Rt. pattern adapts its two-row extension of 8 marked nodes (Co, Do, De, Ae, Ao, Bo, Be, Ce) as the figure. For collision analysis, which essentially need consider both vertical and horizontal positions, a same number mark, such as 08 or Ae, may not identify a same node in both patterns, for easy helix extension presentation; however, such difference can be analyzed via their positions. Both patterns' helix extensions and upper parts are similar to the situation of L-emb.=0.

In Lt. pattern, nodes Co, Do, Ao, Bo, Be, Ce, 11 can keep lag, and nodes 12, 08, 05, 09, 13, 16, 15, 14 (L-emb.>0) can keep lead. Nodes within 13, 09, whose row extension part first, of Rt. pattern are the same as nodes within 14, 10, whose row extension part also first, of Lt. pattern; their differences are sequentially converse in each row and purse extension part. Nodes within 14, 10 can keep lead by definitely adding more nodes.

(2) Consider $x_2 = 1; y_1 - y_2 \geq 2; y_1 = n-2, y_2 = 0$.

The below two basic Lt. and Rt. Hamiltonian path patterns in a SW(4,4) network apparently have no collision except for end nodes.

05,08,04,01,02,03,07,06,10,09,13,16,12,11,15,14 Lt.

05,01,02,03,04,08,12,16,15,11,07,06,10,09,13,14 Rt.

The helix extensions of both patterns start from different columns, and they have consistent vertical-horizontal sequence so that they will not collide. In the upper part, Lt. pattern has 6 nodes, and Rt. pattern has 7 nodes. Nodes 08, 04, 07, 06, 10, 09, 13 of Lt. pattern can keep lag by definitely not-more adding nodes. Nodes 01, 02,03, 16, 12, 11, 15 of Lt. pattern can keep lead by definitely not-less adding nodes. Nodes within 07,06 and 15,14 of Lt. pattern have consistent sequential order as the compared nodes within 07,06 of Rt. pattern and can keep lag by definitely not-more adding nodes.

(3) Consider $x_2 \geq 3$; $m \geq 6$; ($y_1=3, y_2=1$) or ($y_1=2, y_2=0$)

The below two basic Lt. and Rt. Hamiltonian path patterns in a SW(6,4) network apparently have no collision except for end nodes. 01,07,13,14,20,19,24,18,12,06,05,04,03,02,08,09,15, 21,22,23,17,11,10,16 Tp.

01,06,05,04,10,11,17,23,22,21,20,19,24,18,12,07,13, 14,08,02,03,09,15,16 Bt.

Nodes 07, 13, 14, 20, 19, 24, 18, 12, 03, 02, 08, 09, 15 of Tp. pattern keep lag with definitely not-more adding nodes. Nodes 06,05, 04, 10, 11, 17, 23, 22, 21 of Bt. pattern keep lead with definitely not-less adding nodes. Nodes within 03,02 (22,21) of Tp. (Bt.) pattern can keep lead with the compared nodes within 02,03 (08,09) of Bt. (Tp.) pattern because 22,21 (03,02) exist. The common nodes within 02,03 (08,09) of Bt. (Tp.) pattern have consistent sequential order so that they will not collide.

Proof on subcase (d): $y_1 - y_2 \geq 2$, $x_1=0$, x_2 odd, ($y_1 \leq n-2, y_2 \geq 2$) or ($y_1 \leq n-3, y_2 \geq 1$)

The below two sets of basic Lt. and Rt. Hamiltonian path patterns in a SW(4,6) network apparently have no collision except for end nodes.

05,08,04,01,02,03,07,06,10,09,13,17,18,22,21,24, 23,19,20,16,12,11,15,14 Lt.

05,01,02,03,04,08,12,16,20,19,15,11,07,06,10,09, 13,17,21,24,23,22,18,14 Rt. (I)

09,05,01,02,03,04,08,12,11,07,06,10,14,15,19,20, 16,13,17,21,24,23,22,18 Lt.

09,13,16,12,11,07,03,04,08,05,01,02,06,10,14,15, 19,20,24,23,22,21,17,18 Rt. (II)

(1). Consider that both L-embeddings and R-embeddings can exist but x_2 or $m - x_2 \geq 3$, i.e., ($x_2 \geq 1, m - x_2 \geq 3$) or ($x_2 \geq 3, m - x_2 \geq 1$).

On the upper part, the Lt. pattern has 6 nodes, and the Rt. pattern has 7 nodes. Then, helix extensions of two patterns start from different columns, yet having consistent “vertical and then horizontal” order; hence, two helix extensions cannot collide. Nodes 08, 04 and clusters Dx, Dw of Lt. pattern keep lag with definitely not-more adding nodes. Ax, Aw of Lt. pattern will keep lag

after the first embedding. If now upper block exists, 01,02 of Lt. pattern can keep lead. Nodes 01, 02, 03 and clusters Bx, Bw, Cw, Cx of Lt. pattern keep lead with definitely not-less adding nodes.

Nodes within 01,02 of Rt. pattern keep lag but have consistent order with those compared nodes within 01,02 of Lt. pattern. In Rt. pattern, two pure nodes essentially within Bw,Aw per embedding, they will not collide the compared nodes of Lt. pattern. Specifically, considering the closest separation, the first (the second) node within Bw,Aw Rt. pattern has equivalently 3a and 3 (4, respectively) nodes before so that in to the compared node within 01,02 of Lt. pattern is one node-separation; then these two compared nodes of both patterns will be increased by the multiples of ($2a + 2$ nodes) per embedding (including L-emb.) without collision. Similarly, Bw,Cw of Lt. pattern and Bx,Bw of Rt. pattern (both become 02,03 if no upper extension exists) will not collide.

On the lower part, nodes 07, 06, 10,09, 13, 17, 18, 21, 24, 22,23 and By, Bz of Lt. pattern keep lag with definitely not-more adding nodes; node 16, 20, 12, 19, 15, 11 and Dy, Dz, Cz, Cy of Lt. pattern, keep lead with definitely not-less adding nodes. Ay and Az, which can be deemed as special nodes of both patterns, will not be collided for their separation being unable to be divided by four’s multiples. Specifically the difference of total probable adding nodes in both patterns is four’s multiples.

07,06 (15,14) of Rt. pattern, which has nodes within 07,06 of Lt. pattern, can keep lag (lead) and have consistent sequential order as those within Lt. pattern. Similarly, 17,18 of Lt. pattern can keep lag in consistent sequential order to those within Az,21 of Rt. pattern. By enough separation (including clusters) and not-more adding nodes, within nodes of Bw,Aw and 21,22 keep lag to the compared nodes within Az-21 of Rt. pattern. Specifically, the compared part to nodes within 21,22 of Lt. pattern is located the end of Az,21 of Rt. pattern.

(2). Consider the rest of ($y_1 - y_2 \geq 2, x_2 - x_1$ odd), i.e., ($x_2=1, m - x_2 \geq 3$) or ($x_2 \geq 3, m - x_2=1$)

The helix extensions of both patterns start from the same column, but they do not collide because Rt. pattern leads Lt. by two nodes.

On the upper part, nodes 05, 01, 02, 03,04, 08, 06, 10, 14, 15 and clusters Ax, Aw, Bw, Bx, Dw, Dx of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 12,11, 07 and clusters Cx, Cw of Lt. pattern keep lead with definitely not-less adding nodes. On the lower part, nodes 19,20, 17, 21 and clusters Az, Ay of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 16, 13, 24, 23, 22 and clusters Dy, By, Bz,

Cz, Cy of Lt. pattern keep lead with definitely not-less adding nodes. The compared nodes of Dz of Lt. pattern are in both DY, which can keep lag, and Dz, which can keep lead, of Rt. pattern. Similarly, within nodes of DY,CY (Cz,Dz) in Rt.

B. MIHP properties of at-the-same-column classification

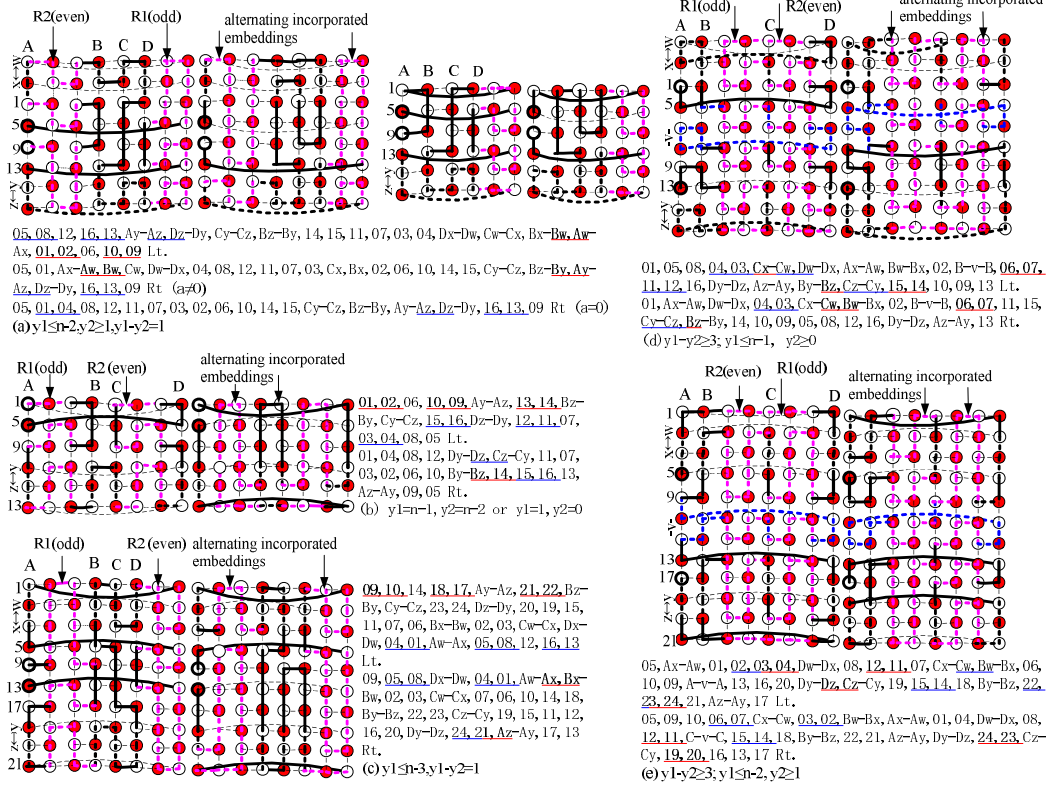


Figure 9. MIHP Properties of At-the-same-column for the Spider-Web Networks.

Proof on subcase (e): $y_1 - y_2 \geq 3, y_1 \leq n-2, y_2 \geq 1$

The below two basic Lt. and Rt. Hamiltonian path patterns in a SW(4,6) network apparently have no collision except for end nodes.

05,01,02,03,04,08,12,11,07,06,10,09,13,16,20,19,15,14,18,22,23,24,21,17 Lt.

05,09,10,06,07,03,02,01,04,08,12,11,15,14,18,22,21,24,23,19,20,16,13,17 Rt.

Both patterns have 12 nodes in the upper part and the part. Both patterns' helix extensions start at the same sequence order but at different position. Both helix extensions have consistent vertical-horizontal order so that they will not collide. For balancing the lead-lag relationships between both patterns, for example, on Cx and 04, alternating incorporated embeddings are considered.

In the upper part, nodes 01, 02, 03, 04, 08, 12, 11 and clusters Ax, Aw, Dw, Dx of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 07, 06, 10, 09 and clusters Cx, Cw, Bw, Bx of Rt. pattern keep lead with definitely not-less adding nodes. Embedded nodes within 02,03 and 03,04 of Lt. pattern cannot collide with the compared nodes within 03,02 and 12,11 of Rt. pattern because the

pattern keep lag (lead) to the compared nodes within Dy-Dz of Lt. pattern. If lower extensions do not exist, 24,23 of Lt. pattern can keep lead.

latter can keep by adding more nodes. Similarly, nodes within 06,07 of Rt. pattern cannot collide with the compared nodes within Cw,Bw of Lt. pattern. Same nodes within both 03,02 of Rt. pattern and Cw,Bw of Lt. pattern, and same nodes within 12,11 of both patterns have consistent sequence order so that they can keep lead/lag relationships after checking the start node whether it can keep lead-lag relationships in the first embedding.

In the lower part, nodes 13, 16, 20, 19 and clusters Dy, Dz, Cz, Cy of Lt. pattern keep lag with definitely not-more adding nodes. Nodes 15, 14, 18, 22, 23, 24, 21 and clusters By, Bz, Cz, Cy of Rt. pattern keep lead with definitely not-less adding nodes. By R1, embedded nodes within 15,14 and within 22,23 of Lt. pattern keep lead with the compared nodes within 15,14 of Rt. pattern. Nodes within 23,24 of Lt. pattern keep lead with nodes within 24,23 of Rt. pattern by definitely more adding nodes. Nodes within 19,20 of Rt. pattern can keep lead with Dz,Cz of Lt. pattern by enough net separation (5 nodes plus 3 clusters) between their start nodes and more per-total-embedding nodes.